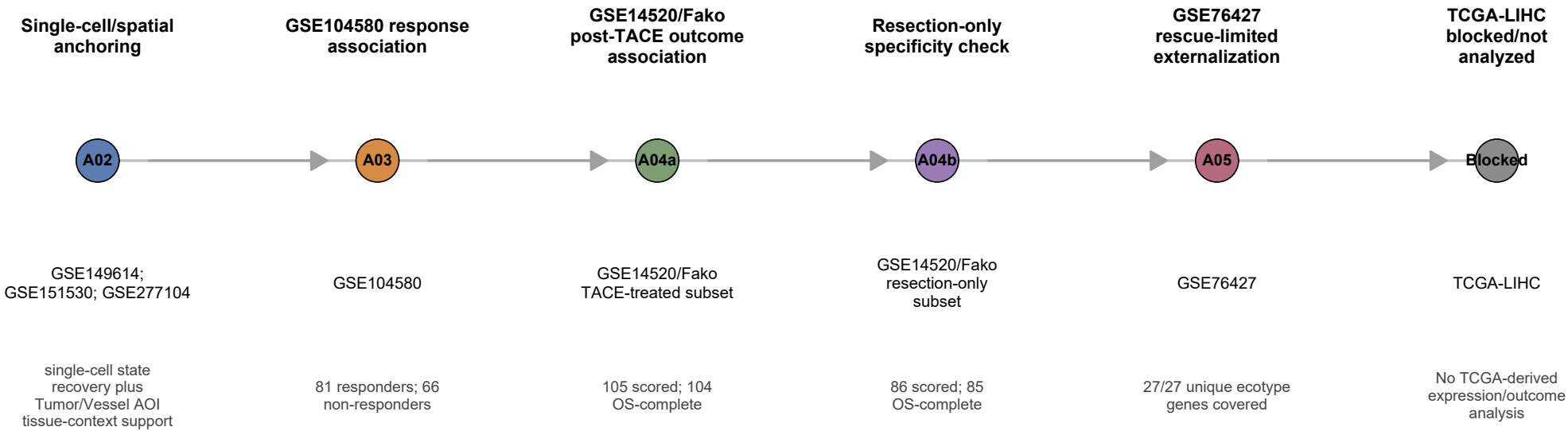


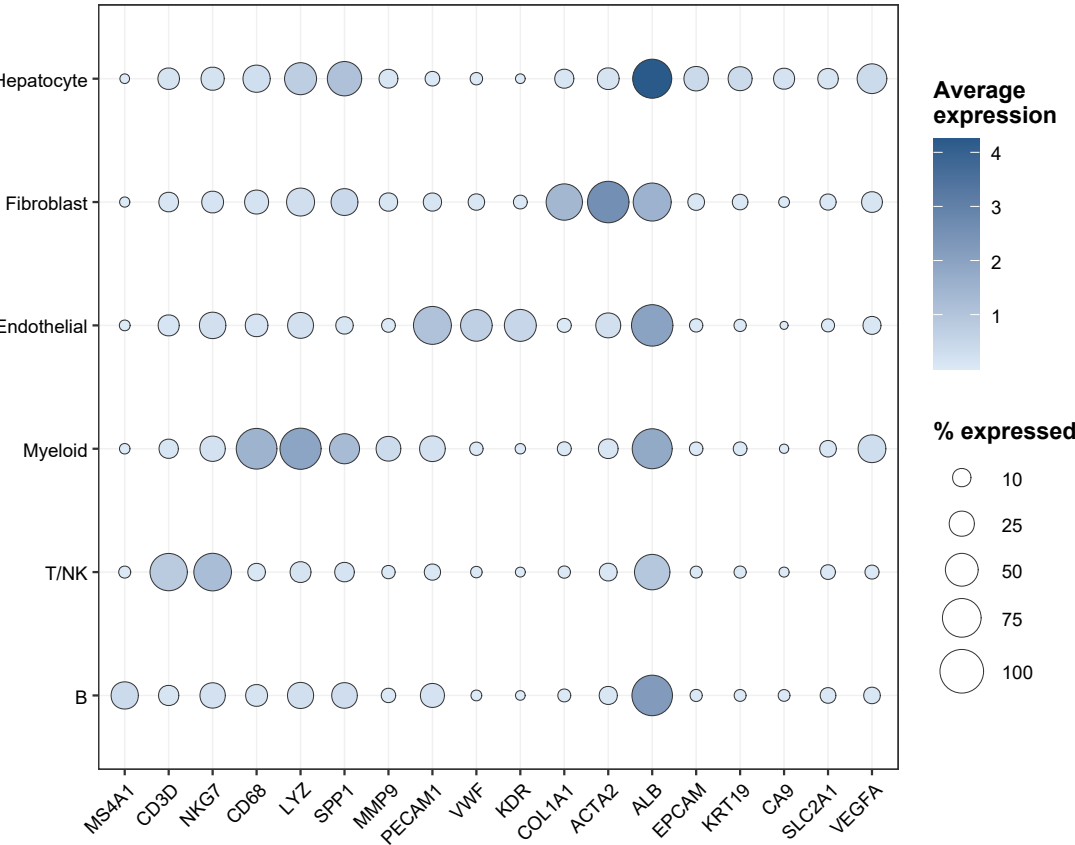
A

Endpoint-first translational association chain
Biological anchoring is separated from endpoint testing and treatment-context outcome assessment



B Marker support across cell types

GSE149614 cell-type-level average expression and percentage expressed



C

Predefined cohort roles and claim-control boundaries

Only GSE104580 is the direct pretreatment response endpoint cohort;
TCGA-LIHC is blocked/not analyzed

